Karthik Balasubramaniam

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Profile

X-ray astrophysicist (Ph.D., Jagiellonian University, 2021) working on spectroscopy and imaging of active galactic nuclei and radio galaxies with Chandra, XMM-Newton and NuSTAR (CIAO, XMM-SAS, XSPEC, relxill). Research involvement since 2014, beginning with instrumentation and M.Sc. projects at the Indian Institute of Astrophysics, Bengaluru, after a B.E. in mechanical engineering. Nine peer-reviewed publications; one first-author paper under review at A&A (arXiv:2610.22168) and a second submitted to arXiv.

Research Experience

RAD@home Astronomy Collaboratory — **Bengaluru, India (remote)**
Remote Collaborator — *Jul 2026 – Present*
X-ray collaborator on the Collimated Synchrotron Thread (CST) radio-galaxy programme (PI Prof. A. Hota): archival Chandra analysis of the cluster environment of CST-2 for a comparison of thermal and radio pressures (ongoing).

X-ray Astrophysics — **Remote; alongside other employment**
Independent Researcher — *Aug 2023 – Present*
Two first-author projects:

- *X-ray point sources towards the giant radio lobes of Centaurus A* (Chandra ACIS-I; CIAO pipeline) — **submitted to A&A**, arXiv:2610.22168 (October 2026).
- *Supermassive black-hole spin from relativistic reflection* (relxill; XMM-Newton and NuSTAR; 57 AGN) — **submitted to arXiv** (October 2026).

National Tsing Hua University — **Hsinchu, Taiwan**
Postdoctoral Research Fellow — *Aug 2022 – Jul 2023*
X-ray spectral analysis of nearby AGN with Prof. Hsiang-Kuang Chang, including Fe K line searches in Chandra data; started the Centaurus A lobe and black-hole spin projects (posters, 2023).

Astronomical Observatory, Jagiellonian University — **Kraków, Poland**
Doctoral Researcher — *Oct 2016 – Oct 2021*
Thesis: *Fluorescent Iron Lines in Various Types of Radio-Loud Active Galactic Nuclei*, under Prof. Łukasz Stawarz (co-advisor Dr. Volodymyr Marchenko). Chandra/XMM-Newton/NuSTAR spectroscopy and imaging; jet–ISM interactions; image deconvolution. Sources: Centaurus A, Pictor A, NGC 3894, CGCG 292-057, 4C+74.26.

Indian Institute of Astrophysics — **Bengaluru, India**
M.Sc. Research Project — *Apr 2015 – 2016*
Search for the unidentified 3.5 keV emission line in X-ray spectra of dwarf galaxies, with Prof. Subinoy Das and Dr. Firoza Sutaria. Reduced archival XMM-Newton data with XMM-SAS and fitted spectra in XSPEC; poster at the ASI meeting (2016).

Indian Institute of Astrophysics — **Bengaluru, India**
Reading Project — *2014*
"Large Scale Structure of the Universe", with Prof. Subinoy Das: used CAMB to examine perturbations in a neutrino-dominated expansion.

Indian Institute of Astrophysics — **Bengaluru, India**
Instrumentation Project (VSIP) — *2014*
With Dr. Padmakar Singh Parihar and Prasanna Deshmukh: mechanical design (SolidWorks) of a dynamic loading assembly for load and performance testing of segmented-mirror-telescope actuators. Co-author of the resulting paper (Deshmukh et al. 2017, *J. Astron. Instrum.*).

Industry & Professional Experience

Ppyrus India Pvt. Ltd. — **Bengaluru, India (remote)**
Innovation & Ecosystem Associate — *May 2026 – Present*

Coordination between academic institutions and technology partners on digital-platform projects; data validation and metadata management.

Wistron Corporation **Hsinchu, Taiwan**
Technical Assistant *Nov 2025 – Feb 2026*
Semiconductor and PCB defect analysis, root-cause troubleshooting, test protocols and defect-traceability documentation.

Vacatronics Technologies **Taiwan**
Maintenance Associate — Cleanroom *Mar 2024 – May 2025*
Class-100 cleanroom maintenance for optical-coating production: facility upkeep and replacement of parts on laser and vacuum coating chambers; process documentation.

HCL Technologies **Poland**
Senior Data Analyst *Jan 2022 – May 2022*
Product-catalogue metadata quality: SQL-based validation, de-duplication and classification.

Restaurant venture **Taiwan**
Investor & Business Partner *2024 – Mar 2026*

Teaching, Mentoring & Outreach

Cram schools (buxiban), Taiwan
STEM & English Instructor (part-time) *2023 – 2026*
Mathematics, science and English at American Eagle (Hsinchu), XL Academy (Taipei) and Little Kangaroo (Taipei).

National Tsing Hua University **Hsinchu, Taiwan**
Workshop Instructor (intermittent) *2022 – 2023*
Workshops on Python-based astronomical data analysis and CIAO/XSPEC spectral fitting.

Jagiellonian University **Kraków, Poland**
Mentor & Teaching Assistant (intermittent) *2016 – 2021*
Mentoring of students in high-energy astrophysics and computational methods.

Jagiellonian Observatory (OAUJ)
Science Outreach & Web Developer *2018 – 2022*
Built and maintained the "Night under the Stars" (NuTS) outreach website (HTML/CSS/JS). www.oa.uj.edu.pl/NUTS

Education

Jagiellonian University **Kraków, Poland**
Ph.D., Astrophysics *2016 – 2021*
Defended Oct 2021. Advisor: Prof. Łukasz Stawarz (Department Head).

Christ University **Bengaluru, India**
M.Sc. Physics (Astrophysics), GPA 7.53 *2014 – 2016*

Visvesvaraya Technological University **India**
B.E. Mechanical Engineering, First Class *2009 – 2013*

Technical Skills

X-ray analysis: XSPEC, relxill, CIAO (Chandra), XMM-SAS, NuSTARDAS, HEASoft

Programming: Python (NumPy, SciPy, Astropy, Pandas, Matplotlib), C/C++, SQL, Bash

Computing: Linux/Unix, Git, Anaconda, Jupyter, MCMC, SLURM/HPC; CAMB (2014)

Plotting / CAD: TOPCAT, Gnuplot, Origin; SolidWorks, ANSYS, Pro-E

Web & docs: HTML/CSS, JavaScript, LaTeX

Professional Development (2026)

be10x AI tools course; generative AI and large language models; Python for data science and machine learning; SQL and data-analytics workflows.

Scholarships, Funding & Patents

- Overseas Fellowship DSTW/ED-1/CR-4/2016-17, Government of Karnataka, India (2016–2021).
- Polish National Science Centre (NCN) Grant 2016/22/E/ST9/00061 (PI: Ł. Stawarz).
- VSIP Fellowship, Indian Institute of Astrophysics (2014).
- Patents: US11668926B2; Taiwan TW109125878, TW110124889 (optical light-guide design).

Peer-reviewed Publications

9 peer-reviewed (7 journal + 2 IAU proceedings) | 66 citations, h-index 4 (Google Scholar, Sep 2026).

- E. Bronzini, ... **K. Balasubramaniam** et al., "Investigating X-ray emission in the GeV-emitting CSOs PKS 1718-649 & TXS 1146+596", *A&A* 684, A65 (2024).
- **K. Balasubramaniam** et al., "X-ray Emission of the γ-ray-Loud Young Radio Galaxy NGC 3894", *ApJ* 922, 84 (2021).
- R. Thimmappa, ... **K. Balasubramaniam** et al., "Complex Structure of the Eastern Lobe of Pictor A", *ApJ* 921, 44 (2021).
- **K. Balasubramaniam** et al., "Chandra View of the LINER-type Nucleus in CGCG 292-057: Ionized Iron Line & Jet–ISM Interactions", *ApJ* 905, 148 (2020).
- R. Thimmappa, ... **K. Balasubramaniam** et al., "Chandra Imaging of the Western Hotspot in Pictor A", *ApJ* 903, 109 (2020).
- G. Bhatta, Ł. Stawarz, A. Markowitz, **K. Balasubramaniam** et al., "Disk–Jet Coupling in the Broad-line Radio Quasar 4C+74.26", *ApJ* 866, 132 (2018).
- P. Deshmukh, P. Parihar, **K. Balasubramaniam** et al., "Dynamic Loading Assembly for Testing Actuators of a Segmented Mirror Telescope", *J. Astron. Instrum.* 6 (2017).
- **K. Balasubramaniam** et al., "Chandra view on the active nucleus of CGCG 292-057: jet–ISM interactions", *Proc. IAU* 342, 222 (2019).
- R. Thimmappa, ... **K. Balasubramaniam** et al., "Preliminary analysis of the X-ray emission from the central regions of Pictor A", *Proc. IAU* 342, 224 (2019).

Submitted Manuscripts

- **K. Balasubramaniam**, H.-K. Chang, Ł. Stawarz, V. Marchenko, "X-ray point sources towards the giant radio lobes of Centaurus A", Submitted to Astronomy & Astrophysics; arXiv:2610.22168.
- **K. Balasubramaniam**, H.-K. Chang, Ł. Stawarz, "What can a uniform reflection pipeline establish about supermassive black hole spin? An audit and a corrected analysis of 57 AGN with XMM and NuSTAR", submitted to arXiv (October 2026).

Selected Talks, Posters & Conferences

- "Searching the Fe line with nearby AGN using Chandra data", Physical Society of Taiwan Annual Meeting (2023).
- "Fluorescent Iron Lines in Radio-Loud AGN", Astronomy Colloquium, NTHU (2022); Extreme Astro Group, U. Michigan (2022).
- Posters: "Black Hole Spins in AGNs: a systematic perspective" & "Exploring Centaurus-A's giant radio lobes", European Astronomical Society 2023, X-ray Universe 2023, ASROC 2023.
- "CSO NGC 3894 with restarting radio jets", 6th Workshop on CSS & GPS Radio Sources (2021).
- "Constraining the accretion-disk structure in 4C+74.26", Cracow School of Theoretical Physics, Zakopane (2019); also IAU Symp. 342, Noto (2018); 20 Years of Chandra, Boston (2019); X-ray Universe, Rome (2017).
- ASI meeting posters: "Unidentified 3.5 keV line in dwarf galaxies" (2016); "Dynamic loading assembly for performance testing of segmented mirror telescope actuator" (with P. Deshmukh and P. Parihar, 2015).